Project Report

Customer Shopping Behavior Analysis

1. Project Overview

The project analyzes customers' shopping behavior using transactional data of 3,900 purchases across various product categories. The main goal of this project is to uncover insights into spending patterns, customer segments, product preferences and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
  - Missing Data (37 values in Review Rating column)

3. Exploratory Data Analysis Using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and .describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previou Purchase
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	1
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	2
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	4
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	3
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
895	3896	40	Female	Hoodie	Clothing	28	Virginia	L	Turquoise	Summer	4.2	No	2-Day Shipping	No	No	3
896	3897	52	Female	Backpack	Accessories	49	Iowa	L	White	Spring	4.5	No	Store Pickup	No	No	4
897	3898	46	Female	Belt	Accessories	33	New Jersey	L	Green	Spring	2.9	No	Standard	No	No	2
898	3899	44	Female	Shoes	Footwear	77	Minnesota	S	Brown	Summer	3.8	No	Express	No	No	2

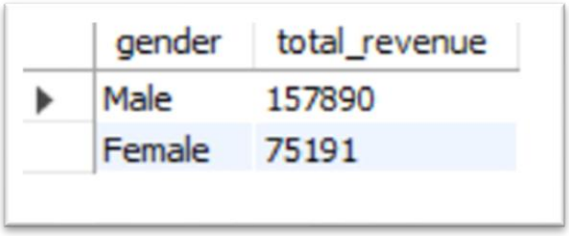
Previous Purchases	Payment Method	Frequency of Purchases
14	Venmo	Fortnightly
2	Cash	Fortnightly
23	Credit Card	Weekly
49	PayPal	Weekly
31	PayPal	Annually
...	...	...
32	Venmo	Weekly
41	Bank Transfer	Bi-Weekly
24	Venmo	Quarterly
24	Venmo	Weekly

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
    - Created age_group column by binning customer ages.
    - Created purchase_frequency_days column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to MYSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

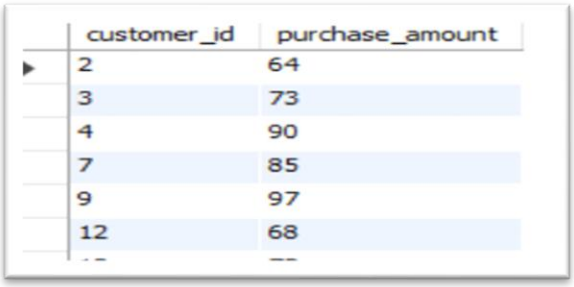
We performed structured analysis in MYSQL to answer key business questions:

- **Revenue by Gender:** Compared total revenue generated by male vs. female customers.



gender	total_revenue
Male	157890
Female	75191

- **High-Spending Discount Users:** Identified customers who used discounts but still spent above the average purchase amount.



customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68

- **Top 5 Products by Rating:** Found products with the highest average review ratings.

Product	Average_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

- **Shipping Type Comparison:** Compared average purchase amounts between Standard and Express shipping.

shipping_type	average_amount
Express	60.4752
Standard	58.4602

- **Subscribers vs. Non-Subscribers:** Compared average spend and total revenue across subscription status.

subscription_status	total_customers	Average_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

- **Discount-Dependent Products:** Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

❖ **Customer Segmentation:** Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	number_of_customers
Loyal	3116
Returning	701
New	83

❖ **Top 3 Products per Category:** Listed the most purchased products within each category.

item_rank	category	item_purchased	total_ord
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145

❖ **Repeat Buyers & Subscriptions:** Checked whether customers with more than 5 purchases are more likely to subscribe.

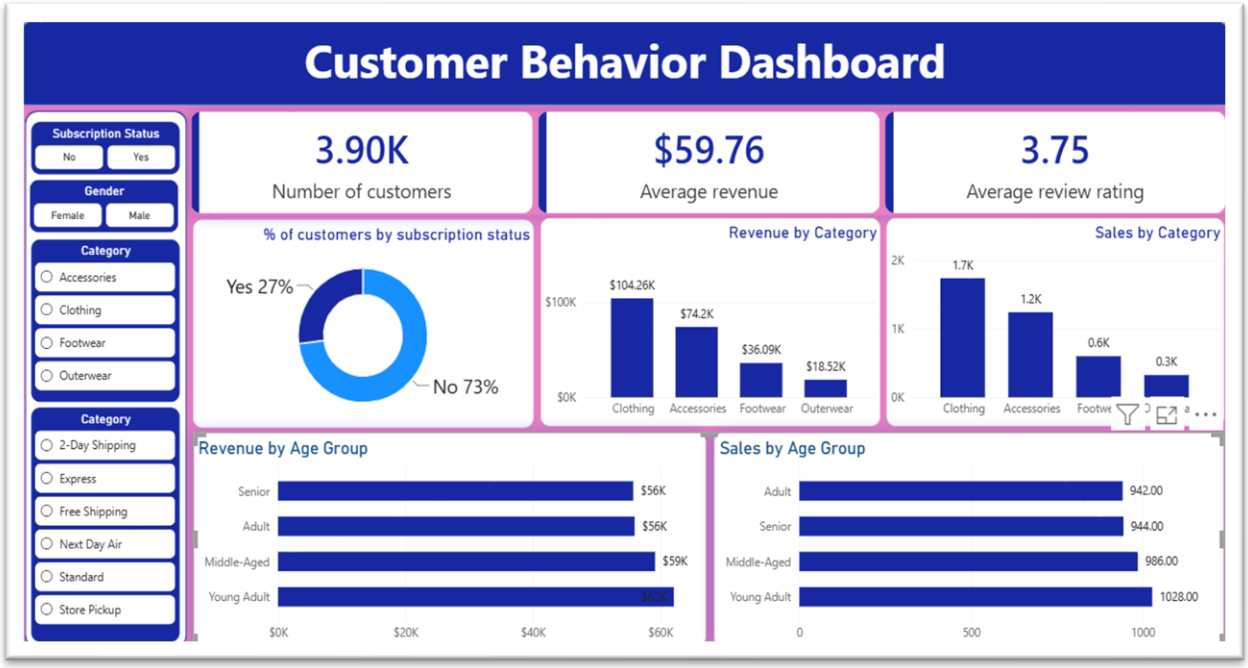
subscription_status	repeat_buyers
Yes	958
No	2518

❖ **Revenue by Age Group:** Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62143
Middle-Aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations:

- **Boost Subscriptions –** Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs:** Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy:** Balance sales boosts with margin control.
- **Product Positioning:** Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing:** Focus efforts on high-revenue age groups and express-shipping users.